

From Scenarios to Consequences: Probabilistic Tsunami Risk Analysis (PTRA) with Geo-INQUIRE services

Lecture Notes for Probabilistic Tsunami Risk Analysis

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Introduction

These notes provide some background and explanation for probabilistic tsunami risk analysis. Probabilistic Tsunami Risk Analysis (PTRA, Behrens et al. 2023, Sørensen et al. 2026) provides a probabilistic estimate of tsunami risk. We can define tsunami risk as *the probability of exceeding a certain value of a risk metric due to a tsunami for people or assets exposed to tsunami risk in a specific location in a given window of time*. In another word, risk assessment entails responding to several questions: **who** or **what** is at risk? And **where** are they located? **What** is/are our risk metric? **Which** is the window of time for which we would like to assess the risk? **What** do we mean by a significant tsunami event? **How** is the intensity of a tsunami measured? **How** do we assess this probability? In this short lecture note, we provide some basic information above the above questions. Wherever possible, we point you towards relevant literature and more in-depth reading. Although in this document we are focusing on the tsunami risk assessment, the document can be generalised to natural hazards like earthquakes, floods, hurricanes, landslides, ..., etc. This lecture note is focused on earthquake-induced tsunami events, however, the concepts described are quite general to all types of tsunami events. This lecture note assumes an understanding of probabilistic concepts for the purpose of hazard and risk modelling. You can use the following [lecture series on fundamentals of probability and applied statistics](#) to gain a basic understanding of the underlying probabilistic concepts for PTRA.

Who or what is at risk?

It is evident that most central element of risk analysis is the definition of people or assets at risk. This entails defining the people or assets (e.g., buildings, infrastructure, crops) who may be potentially exposed to tsunami risk. This concept is more formally defined as *risk exposure* (see Pittore et al. 2017). How do we define who or what is exposed to tsunami risk? One possible way would be to identify the coastal area which would be potentially affected by tsunami and then evaluate who is living in that area and what type of physical assets are in that area. Such an assessment of exposed people and assets is referred to as exposure analysis or exposure modelling. For further reading about exposure modelling, you can refer to the [Exposure modelling chapter](#) within the [PTRA Cookbook](#). Generally, exposure can be classified into two macro categories: (1) human exposure (people at risk); (2) physical exposure (“things” at risk). Throughout these notes, we also follow these two categories.

Where are they located?

In the previous point, we saw that, within the context of tsunami risk assessment, the question of who or what is at risk and where are they located are inherently related to each other. This is more generally true for all location-dependent hazards. Indeed, it is possible to define the “*hazard footprint*” for a location-dependent hazard. This entails the areal extent which is affected by the hazard in question.

In fact, exposure modelling tackles both questions of who and what is exposed to risk and where they are located. One may ask, if our purpose is to do a probabilistic analysis, how can

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we establish with certainty the hazard footprint? In fact, delineation of a hazard footprint is also subjected to uncertainties and will depend on the plausible tsunami scenario we are discussing. Therefore, one way to define the tsunami risk exposure is to define a maximum envelope of the coastal area at risk or even define the administrative area of reference (e.g. a village, city, province).

The exposure model discretisation

One of the main parameters affecting the risk analysis is the discretisation of the exposure model. In simple words, this addresses the smallest spatial/areal unit of interest containing people or things exposed to risk. This is also referred to as the *resolution* of the exposure model. For example, we can define Voronoi cells (e.g. Pittore et al. 2020), spatial grids or pixels (e.g. Woods et al. 2025), administrative boundaries (Crowley et al. 2021), or an asset-level resolution (e.g., building-to-building resolution, Schorlemmer et al. 2026). It is important to note that the exposure model defines the underlying risk model set-up; in other words, the resolution of the exposure model defines and sets the resolution of the risk results.

What is our risk metric?

As the names suggest, the risk metric is the parameter or the set of parameters we use for measuring risk. Therefore, risk metrics measure the losses to people, things, or businesses due to a disruption. In a very simple and straightforward manner, the risk metric categories were referred to as the “3D”s: Deaths, Dollars, Down-time. In some risk assessment frameworks, the risk metric is also defined as a “Decision Variable” (Porter 2003, Deierlein and Moehle, 2004). This definition is a direct nod to the fact that the risk metrics provide the basis for “risk-informed” or “evidence-based” decision making. Considering the probabilistic nature of risk assessment, we usually report some statistical property of the risk metric in question. For example, we can report the expected value of the risk metric in each window of time. For example, a common risk metric representation is the annual expected losses (AEL), also known as the average annual losses (AAL). Another common representation is the mean annual rate of exceeding a given level of losses, also known as the loss curve (in case of economic losses) and the mean annual rate of exceeding a certain number of casualties known also as the fatality curve (in case of loss of life).

Damage metrics: Damage scale and damage states

In the case of physical damage to assets, we usually use discrete physical damage states to discretise physical damage. You can refer to Jalayer et al. (2023) for a detailed description of the damage different damage scales. Damage scale is the ensemble of *mutually exclusive and collectively exhaustive (MECE, see Lecture Nore 1 for the definition of MECE) damage states* (DS_i) used to describe the *whole* range of possibilities in terms of damage for an exposed asset.

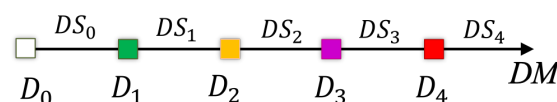


Figure 1: A physical damage scale containing the ensemble of damage levels and damage states

As seen in Fig. 1, the damage levels D_i mark the thresholds of the damage states.

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Which is the window of time for which we would like to assess the risk?

As we saw in the definition above, any risk assessment needs to define a window of time for which risk is being assessed. The window of time of reference varies based on many different parameters such as, (1) the hazard in question; (2) the decision-making framework. For example, some hazards are time-varying, and we would need an appropriate reference time window which would capture the time variability; examples are the aftershock hazard which has a significant variation in the first few days after a large earthquake. In such as case, a 24-hour time window may seem appropriate. As time passes, the rate of variation of the seismicity may decrease and a 1-week time window may be more appropriate. The choice of the time window also depends on the decision-making context. For example, for designing buildings and physical infrastructure generally a window of 30 to 50 years is chosen, based on the designated nominal lifetime of the infrastructure. For reporting losses for insurance and re-insurance industry decision-making context, a one-year window is usually chosen.

How the intensity of a tsunami is measured?

We usually represent the intensity of a tsunami event through a variable called the intensity measure (IM). The intensity measure (IM) is a scalar or vector that plays the role of interface variable between hazard and vulnerability. Examples of intensity measures for earthquake are PGA and Spectral Acceleration. Examples of intensity measures for Tsunami are wave amplitude, flow depth, current velocity, momentum flux, or maximum inundation height. For a given tsunami scenario, the *inundation footprint* of the tsunami can be shown as the map of the IM values for each exposed asset cell grid/unit.

What do we mean by a significant tsunami event?

Not all tsunami events will be of our interest for the purpose of risk calculations. We are usually interested in a subset of all the tsunami events which have significant potential for causing losses to human life, things, and businesses. We know that tsunamis are always triggered from another event or a set of cascading events. One way to define tsunami events of the interest is through the magnitude of the primary event which is triggering the tsunami. For earthquake-induced tsunami events (which are the focus of this lecture note), the tsunami events of interest are defined as those which are triggered by an earthquake with magnitude larger than a specified minimum magnitude denoted as M_{min} .

How do we assess risk?

This lecture note is pivoting around this question. Within the space of probabilistic and quantitative risk assessment approaches, there are different frameworks available for calculation of risk. PTRA is performed within a forward probabilistic framework, in which a tsunami leads to direct consequences in terms of loss of life and property (Fig. 2). As you can see in the figure, this framework has a consequential structure. The figure also demonstrates that the estimated risk metrics form a basis for risk-informed decision-making. You can make an extra reading of the different frameworks available for PTRA here (The Cookbook: [PTRA Framework Chapter](#)).

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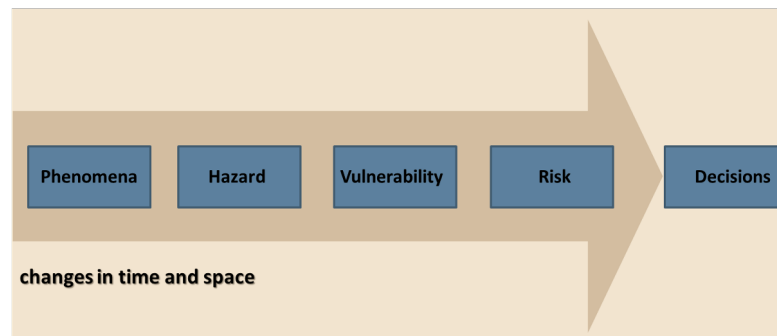


Figure 2: A forward framework for risk-informed decision making

The structure of this lecture note

In the first section after introduction, we are going to discuss two different workflows for PTRA. In the following sections we are going to describe in a detailed and step-by-step manner how each workflow works.

The PRTA Workflows

This lecture note is focused on PTRA based on time-invariant risk analysis. This means that we assume that the occurrence of the consequences such as economic loss or loss of life are described by Homogenous Poisson Processes (see [Lecture Note 5](#)). Therefore, time-variant risk analysis is out of the scope of this lecture note. We can classify the PTRA workflows into two main categories:

Simulation-based PTRA (also referred to as the event-based PTRA)

This approach follows a classic Monte Carlo simulation workflow –or any other simulation scheme from the Monte Carlo family such as Markov Chain Monte Carlo (MCMC) or Importance Sampling. This workflow is more suitable for regional risk assessment (e.g., for spatially distributed assets).

We are going to go over two different variations of this workflow. In the first variation, to simplify the workflow, we start from already generated tsunami inundation scenarios for PTHA. In this variation, each tsunami scenarios is associated with a rate of occurrence. This is similar to the Open Quake workflow for probabilistic seismic hazard analysis (PSHA) described in Silva et al. 2014. It is also described in the Cook Book [Frameworks Chapter](#) under the code FW1A.

- **The calculation pathway for this workflow is:**
Hazard scenarios → Vulnerability → Risk (with exposure being the “container” defining the elements at risk)

The second variation is fully simulation-based; that is, we do not start from already generated tsunami inundation scenarios. Rather, we simulate each scenario “on-the-fly” all the way towards the consequences. This latter variation is fully aligned with the Monte Carlo Simulation Scheme and lends itself readily to adaptive importance sampling or adaptive Markov Chain Monte Carlo schemes.

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- **The calculation pathway for this workflow is:**

Source selection → Rupture generation → Tsunami generation → Tsunami propagation and inundation → Hazard scenarios → Vulnerability → Risk (with exposure being the “container” defining the elements at risk)

In this workshop, we focus on the first variation –as the name of the workshop indicates.

PTRA using hazard curve

In this workflow, our starting point is the tsunami hazard curve (output of PTHA). We have two possibilities for risk calculations here: 1) direct numerical integration; 2) analytical closed-form approximation. This workflow is more suitable for single assets at risk (a single infrastructure). This workflow is described in the Cook Book [Frameworks Chapter](#) under the code FW1B.

- **The calculation pathway for this workflow is:**

Hazard curves → Vulnerability → Risk (with exposure being the “container” defining the elements at risk)

Simulation-based PTRA

There are alternative ways to describe the simulation-based PTRA procedure. We first describe the first variation as the method adopted in this workshop –considering that the starting point of our calculations are the set of plausible tsunami scenarios already generated for probabilistic tsunami hazard analysis (PTHA) calculations. Then we describe the second variation, including more details about the scenario generation procedure. As we mentioned before, the simulation-based PTRA workflow is more suitable for spatial risk assessment; that is, when the people and assets at risk are distributed over a spatial extension.

Let us imagine that A denotes the coastal area of reference for which we would like to calculate the tsunami risk. A can be defined as the zone susceptible to tsunami risk; for example, the minimum of the altitude above sea-level less than 100 meters and within a 1 km coastal stripe. Let $C = \{c_k; k = 1: N_{cell}\}$ be the set of exposure cells (i.e. the cells containing exposed assets) containing N_{cell} cell elements representing the exposed assets. For example, c_k can be Voronoi cells, grid cells, administrative unit boundaries, or individual assets (such as buildings, bridges, green areas).

Losses due to economic damage: In this lecture note we focus on direct economic losses. This means that we estimate the economic losses as the losses incurred to repair the damage caused by tsunami to the asset in question. Let us denote by RC_k the direct losses associate with c_k due to a tsunami event of interest. We can express the total losses as the sum of the losses incurred within each cell:

$$L = \sum_{k=1}^{N_{cell}} RC_k = \sum_{k=1}^{N_{cell}} LR_k \cdot R_k \quad (1)$$

This is quite straightforward; the total sum incurred to the assets within the area of reference is equal to the sum of the individual losses associated with each cell containing exposed assets. Note that the repair costs are usually normalised with respect to the replacement cost for the

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physical asset in question. It is common to work with loss ratio which is equal to the repair costs divided by the cost of replacement of the assets, therefore $RC_k = LR_k \cdot R_k$ where R_k is the cost of replacement for the asset located in cell c_k and LR_k is the normalised loss ratio varying between zero and one.

Loss of life: Let us imagine that the population density in grid cell c_k is equal to N_k , that is, N_k people are in grid cell c_k . The total number of casualties incurred will be equal to:

$$L = \sum_{k=1}^{N_{cell}} X_{F,k} N_k \quad (2)$$

Where $X_{F,k}$ is a Bernoulli random variable (see [Lecture Note 2](#)) which is equal to unity in case of loss of life and is equal to zero in case of opposite.

PTRA starting from generated inundation scenarios

As mentioned above, the simulation-based PTRA is framed within a Monte Carlo Simulation formulation. Let us denote the risk metric as variable L (where L reminds us of losses generically). We can calculate the mean annual rate of L exceeding a specified value l as follows (see [Lecture Note 1](#) and [Lecture Note 2](#) for the probabilistic notation and the types of probability distributions; [Lecture Note 5](#) and [Lecture Note 6](#) related to Poisson Distribution and the filtered Poisson Distribution):

$$\lambda(L > l) = \sum_{i=1}^{N_{source}} \sum_{\mathbf{e}_{ij} \in E_i} P(L > l | \mathbf{e}_{ij}) \lambda(\mathbf{e}_{ij}) \quad (3)$$

Where N_{source} denotes the number of tsunamigenic seismic sources considered in the study and E_i refers to the set of plausible tsunami scenarios which can be generated from source i ; $\mathbf{e}_{ij}$ refers to tsunami scenario j generated from source i . $P(L > l | \mathbf{e}_{ij})$ refers to the probability that the risk metric L exceeds a specific value l ; and $\lambda(\mathbf{e}_{ij})$ denotes the mean annual rate of occurrence of a specific scenario $\mathbf{e}_{ij}$. A tsunami scenario is described by the magnitude M of the tsunamigenic event and a series of other source parameters defined as vector $\mathbf{s}$.

In a fully simulation-based framework (more suitable for time-variant risk analysis, see Papadopoulos et al. 2021 for an example), the tsunamigenic earthquake itself (e.g., the earthquakes with different source characteristics), can be stochastically generated in time to reflect $\lambda(\mathbf{e}_{ij})$. Herein, $\lambda(\mathbf{e}_{ij})$ is pre-calculated and only the part $P(L > l | \mathbf{e}_{ij})$ is evaluated using the Monte Carlo Simulation. You can refer to Basili et al. (2021) for a detailed description of how $\lambda(\mathbf{e}_{ij})$ is calculated. Herein, we use $\lambda(\mathbf{e}_{ij})$ and E_i information already provided for a PTHA workflow. This gives us with the possibility of focusing more closely on the risk-related components such as exposure and vulnerability.

Ignoring the uncertainties involved in tsunami modelling and propagation, we can assume that the rate of occurrence $\lambda(\mathbf{e}_{ij})$ will also be associated to the corresponding inundation footprint of the tsunami. Suppose that we show the inundation footprint for scenario $\mathbf{e}_{ij}$ at cell c_k as $\mathbf{im}_{ij} = [im_{ijk}, k = 1: N_{cell}]$; that im_{ijk} is the value of the intensity (e.g. maximum flow depth) at cell c_k . We can show that $\lambda(L > l)$ can be calculated as:

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$$\lambda(L > l) = \sum_{i=1}^{N_{source}} \sum_{e_{ij} \in E_i} P(L > l | \mathbf{im}_{ij}) \lambda(\mathbf{e}_{ij}) \quad (4)$$

Considering epistemic uncertainties in rate of occurrence of scenarios

Now imagine that we would like to consider epistemic uncertainties related to the rate of occurrence of scenarios. That is, instead of $\lambda(\mathbf{e}_{ij})$ we consider its probability distribution $p(\lambda(\mathbf{e}_{ij}))$. In that case, we can write Equation (3) as follows:

$$p(\lambda(L > l)) = \sum_{i=1}^{N_{source}} \sum_{e_{ij} \in E_i} P(L > l | \mathbf{im}_{ij}) p(\lambda(\mathbf{e}_{ij})) \quad (5)$$

Since we are calculating the mean annual rate of exceeding a loss value through simulation, this means that for each rate value sampled from probability distributions $p(\lambda(\mathbf{e}_{ij}))$, we will have the corresponding realisation of the loss curve. This way, we can propagate the epistemic uncertainties in the estimation of the rates to the level of the losses.

The case of physical exposure:

Imagine we use a physical damage scale consisted of MECE damage states $DS = \{DS_l: l = 1: N_{DS}\}$. We can simulate a vector of random damage states for each of the cells c_k by using the fragility function. The fragility model can be defined as the PMF (Probability Mass Function, see [Lecture Note 2](#) on the discrete probability distributions) for the damage states denoted as $P(DS_l | im = x), l = 1: N_{DS}$. Therefore, using the Fragility model, we can simulate a damage state for each im_{ijk} value corresponding to cell c_k related to the inundation footprint $\mathbf{im}_{ij}$ for scenario e_{ij} . According, we can call the vector of damage states simulated for scenario $\mathbf{e}_{ij}$ as $\mathbf{DS}_{ij} = [DS_{ijk}, k = 1: N_{cell}]$. We can use the total probability theorem (see [Lecture Note 1](#)) to expand with respect to all the damage states within the damage scale:

$$p(\lambda(L > l)) = \sum_{i=1}^{N_{source}} \sum_{e_{ij} \in E_i} \sum_{n=1}^{N_{DS}} P(L > l | \mathbf{DS}_{ij}) P(\mathbf{DS}_{ij} | \mathbf{im}_{ij}) p(\lambda(\mathbf{e}_{ij})) \quad (6)$$

It is important to note that the fragility models are defined for specific types of physical asset distinguished by strings which are called taxonomy. When our exposure model has resolution at the asset level (e.g. The Global Dynamic Exposure model), this is more straightforward since each unit of exposure is corresponding to a specific typology (and hence taxonomy). Things get a bit more complicated when the exposure model has an aerial resolution (e.g., Voronoi cell, grid cell). In this case more than one asset could be located inside the cell and they could correspond to different typologies and fragility models. There are different ways to resolve this. For simplicity let us assume that in such a case we would use an average fragility model which is equal to the weighted (e.g. proportional to the frequency of each typology) average of the different typologies located within the cell in question. The alternative is to simulate the building class occupying cell c_k based on the reported frequency of each building class within the cell. This is quite straightforward since physical exposure models usually associate relative frequencies or probabilities to each building class. These probabilities can be used to build a probability mass function (PMF) for different building classes and sample the building class from this PMF. To avoid making the formulation too complicated, we have not shown this explicitly in the formulation above.

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Based on the vector of the simulated damage states $\mathbf{DS}_{ij}$, we can simulate the vector of repair costs for all the exposure cells C . This will be done through the cost models or consequence models represented by a conditional PDF (Probability Density Function, see [Lecture Note 3](#)) denoted as $f_{RC|DS}(rc)$ which describes the conditional distribution of losses RC for each exposure cell unit given the damage state DS . We can use this PDF to simulate a vector of repair cost values for scenario $\mathbf{e}_{ij}$ and vector of damage states $\mathbf{DS}_{ij}$ denoted as $\mathbf{RC}_{ij} = [RC_{ijk}, k = 1:N_{cell}]$. Therefore, the mean annual frequency of exceeding a given loss value can be calculated as follows:

$$p(\lambda(L > l)) = \sum_{i=1}^{N_{source}} \sum_{\mathbf{e}_{ij} \in E_i} \sum_{n=1}^{N_{DS}} \int_{\Omega_{RC}} P(L > l | \mathbf{RC}_{ij}) f(\mathbf{RC}_{ij} | \mathbf{DS}_{ij}) P(\mathbf{DS}_{ij} | \mathbf{im}_{ij}) d\mathbf{RC}_{ij} p(\lambda(\mathbf{e}_{ij})) \quad (7)$$

Where $P(L > l | \mathbf{RC}_{ij})$ is the probability that loss L exceeds l given that the repair costs for all the assets are known and Ω_{RC} is the domain of loss vector $\mathbf{RC}_{ij}$.

Here we use the Monte Carlo Simulation to solve the multi-dimensional numerical integral in Equation (7). Consider that we use $m = 1:N_\lambda$ realisations of the rate of occurrence of each scenario for representing the epistemic uncertainties in the rates and N_{sim} simulations for estimation of $\lambda_m(L > l)$, the m^{th} realisation of $\lambda(L > l)$ can be approximated using Monte Carlo simulation (see [Lecture Note 9](#)):

$$\lambda_m(L > l) \cong \frac{1}{N_{sim}} \sum_{i=1}^{N_{source}} \sum_{\mathbf{e}_{ij} \in E_i} \mathbf{1}(L_{ij} > l | \mathbf{e}_{ij}, \mathbf{im}_{ij}, \mathbf{DS}_{ij}, \mathbf{RC}_{ij}) \lambda_m(\mathbf{e}_{ij}) \quad (8)$$

Where N_{sim} is equal to the total number of scenarios (number of sources times the number of scenarios generated for each source), $\mathbf{1}(L_{ij} > l | \mathbf{e}_{ij}, \mathbf{im}_{ij}, \mathbf{DS}_{ij}, \mathbf{RC}_{ij})$ is the unit function: it is equal to unity if $L = \sum_{k=1}^{N_{cell}} RC_k$ is larger than l and zero otherwise, and $\lambda_m(\mathbf{e}_{ij})$ is the m^{th} realisation of the rate of occurrence of scenario $\mathbf{e}_{ij}$ sampled from probability distribution $p(\lambda(\mathbf{e}_{ij}))$. Note that this provides us with N_λ realisations of the loss curve $\lambda_m(L > l)$. We can report the median, mean, 16th, and 84th percentiles of these curves, providing a confidence band which reflects the propagation of epistemic uncertainties in the estimation of the rate of occurrence of each tsunami scenario.

The case of human exposure:

Imagine that we represent the event of loss of life with a logical statement F , where $F \equiv \text{loss of life}$ and its opposition $\bar{F} \equiv \text{no loss of life}$. The *fatality function* can be defined as the conditional probability of loss of life given a specific value x of intensity measure im denoted by $P(F | im = x)$. We can use a Bernoulli random variable X_F which assumes the value of 1 with probability $P(F | im = x)$ in case of loss of life and value of 0 with probability $1 - P(F | im = x)$ in case of no loss of life. For each of the cells c_k by using the fatality function, we can simulate the Bernoulli random variable $X_{F,ijk}$ for each im_{ijk} value belonging to the inundation footprint $\mathbf{im}_{ij}$ for scenario $\mathbf{e}_{ij}$. Accordingly, we can call the vector of fatality information simulated for scenario $\mathbf{e}_{ij}$ as $\mathbf{X}_{F,ij} = [X_{F,ijk}, k = 1:N_{cell}]$. We can use the total probability theorem (see the lecture notes provided) to expand with respect to the two states of the Fatality (1 and 0 or F and $\bar{F}$):

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$$p(\lambda(L > l)) = \sum_{i=1}^{N_{source}} \sum_{\forall \mathbf{e}_{ij} \in E_i} \sum_{n=1}^2 P(L > l | \mathbf{X}_{F,ij}) P(\mathbf{F} | \mathbf{im}_{ij}) p(\lambda(\mathbf{e}_{ij})) \quad (9)$$

Where the generic loss variable is the total number of fatalities $L = \sum_{k=1}^{N_{cell}} N_k X_{F,ijk}$ where N_k is the population in cell grid k . Note that in this case, instead of summing over number of damage states, we sum over the only 2 possible states of fatality (1 and 0 or F and $\bar{F}$).

Here again we use the Monte Carlo Simulation to solve the multi-dimensional numerical integral in Equation (9). Consider that we use $m = 1: N_\lambda$ realisations of the rate of occurrence of each scenario for representing the epistemic uncertainties in the rates and N_{sim} simulations for estimation of $\lambda_k(L > l)$, the m^{th} realisation of $\lambda(L > l)$ can be approximated using Monte Carlo simulation:

$$\lambda_m(L > l) \cong \frac{1}{N_{sim}} \sum_{i=1}^{N_{source}} \sum_{\forall \mathbf{e}_{ij} \in E_i} \mathbf{1}(L_{ij} > l | \mathbf{e}_{ij}, \mathbf{im}_{ij}, \mathbf{X}_{F,ij}) \lambda_m(\mathbf{e}_{ij}) \quad (10)$$

Where N_{sim} is equal to the total number of scenarios (number of sources times the number of scenarios generated for each source) and $\lambda_m(\mathbf{e}_{ij})$ is the m^{th} realisation of the rate of occurrence of scenario $\mathbf{e}_{ij}$ sampled from probability distribution $p(\lambda(\mathbf{e}_{ij}))$. The same as above, this provides us with N_λ realisations of the fatality curve $\lambda_m(L > l)$. We can report the median, mean, 16th, and 84th percentiles of these curves, providing a confidence band which reflect the propagation of epistemic uncertainties in the estimation of the rate of occurrence of each tsunami scenario.

The general formulation of PTRA using filtered Poisson

Here we describe an alternative formulation of PTRA which uses the properties of a filtered Poisson distribution (see [Lecture Note 6](#)). The main difference with the previous formulation is that here we estimate the rate of occurrence of tsunami events of interest for each seismic source. In this approach we do not need to associate the rate of occurrence to each tsunami scenario. Also, here the simulation-based PTRA is framed within a Monte Carlo Simulation formulation. We can calculate the mean annual rate of L exceeding a specified value l as follows:

$$\lambda(L > l) = \sum_{i=1}^{N_{source}} \sum_{\forall \mathbf{e}_{ij}} P(L > l | \mathbf{e}_{ij}) \delta(\mathbf{e}_{ij} | \mathbf{s}_i) P(\mathbf{s}_i) \lambda_i(M > M_{min}) \quad (11)$$

Where $\mathbf{s}_i$ is the set of uncertain tsunami parameters (e.g., magnitude, rupture nucleation point, rupture dimensions, slip distribution, friction parameter, Froude number, ...) used for creating tsunami inundation scenarios generated from source i . Consider that $\mathbf{s}_i$ is the set of parameters required for tsunami genesis, propagation, and inundation. In other words, for tsunami realisation j from source i denoted as $\mathbf{e}_{ij}$, we can simulate the inundation footprint $\mathbf{im}_{ij} = [im_{ijk}, k = 1: N_{cell}]$. Compared to the scenario-based formulation in the previous section, instead of going over a pre-defined set of tsunami scenarios, we simulate each scenario “*on-the-fly*”. This is why this variation lends itself well to adaptive simulation schemes such as the adaptive importance sampling or adaptive Markov Chain Monte Carlo. Note that $\delta(\mathbf{e}_{ij} | \mathbf{s}_i)$ is a Dirac Delta function showing that if the set of tsunami parameters $\mathbf{s}_i$ are known, the tsunami scenario $\mathbf{e}_{ij}$ will be also fully known—without uncertainties.

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To solve for $\lambda(L > l)$, we randomly select a seismic source based on the PMF (probability mass function) defined as follows for source i :

$$P(source_i) = \frac{\lambda_i(M > M_{min})}{\sum_{k=1}^{N_{source}} \lambda_k(M > M_{min})} \quad (12)$$

In the next step, depending on how the seismic source is modelled (e.g. as areal source zone or as finite fault) we simulate the random source parameters $\mathbf{s}_i$ from their joint probability distribution $P(\mathbf{s}_i)$. For example, if we are using a finite fault source modelling approach, we simulate the earthquake magnitude ($M > M_{min}$) from its probability density function, then we simulate the point of rupture nucleation along the fault plane. Given the earthquake magnitude, we can simulate the rupture dimensions from the magnitude scaling relationships, we can also simulate the slip distribution along the fault. Once all the components of vector of tsunami parameters are sampled from the joint PMF $P(\mathbf{s}_i)$, we can create the tsunami scenario $\mathbf{e}_{ij}$ and obtain the inundation footprint. The rest is the same as previous section.

For the case of physical exposure, the probability distribution for the mean annual rate of exceedance $\lambda(L > l)$ can be calculated as:

$$p(\lambda(L > l)) = \sum_{i=1}^{N_{source}} \sum_{\mathbf{e}_{ij}} \sum_{n=1}^{N_{DS}} \int_{\Omega_{RC}} \mathbf{P}(L > l | \mathbf{RC}_{ij}) f(\mathbf{RC}_{ij} | \mathbf{DS}_{ij}) P(\mathbf{DS}_{ij} | \mathbf{im}_{ij}) \delta(\mathbf{e}_{ij} | \mathbf{s}_i) P(\mathbf{s}_i) d\mathbf{RC}_{ij} p(\lambda(M > M_{min})) \quad (13)$$

For the case of human exposure, the probability distribution for the mean annual rate of exceedance $\lambda(L > l)$ can be calculated as:

$$p(\lambda(L > l)) = \sum_{i=1}^{N_{source}} \sum_{\mathbf{e}_{ij}} \sum_{n=1}^2 P(L > l | \mathbf{X}_{F,ij}) P(\mathbf{F} | \mathbf{im}_{ij}) \delta(\mathbf{e}_{ij} | \mathbf{s}_i) P(\mathbf{s}_i) p(\lambda(M > M_{min})) \quad (14)$$

The Expected Annual Loss (AEL)

The expected annual loss (EAL) is commonly used as a risk metric. Here, we show that it is equal to the area under the curve $\lambda(L > l)$. We can calculate the area under the loss curve $\lambda(L > l)$ as follows:

$$\begin{aligned} Area &= \int_0^{+\infty} \lambda(L > l) dl = \int_0^{+\infty} l |d\lambda(L > l)| = \lambda(EQ) \int_0^{+\infty} l |dP(L > l | EQ)| = \\ &\lambda(EQ) \int_0^{+\infty} l f_l(l | EQ) dl = E(l | EQ) \lambda(EQ) = EAL \end{aligned} \quad (15)$$

We can use integration by parts to calculate the area under curve as the product of the loss by the absolute value of the increment of the loss curve. We can write the loss curve as the product of mean annual rate of occurrence of a tsunamigenic earthquake of interest $\lambda(EQ)$ and the probability that losses exceed a certain level l given that a tsunamigenic earthquake of interest has taken place denoted as $P(L > l | EQ)$. This is the complementary cumulative distribution function (CCDF) of loss given that a tsunamigenic earthquake of interest has taken place and is denoted as $P(L > l | EQ)$. As you can see in [Lecture Note 3](#), the absolute value of the derivative of the CCDF is equal to the probability density function of loss $f_l(l | EQ)$. Moreover, as you can see in [Lecture Note 4](#), the expected value of losses given that a tsunamigenic earthquake of interest has taken place is equal to the integral of loss multiplied by its PDF. We can argue that

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the product of mean annual rate of occurrence of tsunamigenic earthquake events $\lambda(EQ)$ and the expected value of losses given that a tsunamigenic event of interest has taken place $E(l|EQ)$ is equal to the expected value of annual losses (EAL).

PTRA using the hazard curves

This workflow is suitable for point-by-point risk assessment. Therefore, we describe this formulation for only one cell grid.

In the case of physical exposure, we can calculate the mean annual rate $\lambda(L > l)$ of exceeding loss value l as following:

$$\lambda(L > l) = \int_{\Omega_{RC}} \sum_{n=1}^{N_{DS}} \int_0^{\infty} \mathbf{1}(L > l|RC) f(RC|DS) P(DS|im) dRC |d\lambda(IM > im)| \quad (16)$$

Note that for one asset, the direct economic loss is equal to the repair costs or the product of the loss ratio and the replacement cost for the asset in question. $L = RC = LR \cdot R$. $\lambda(IM > im)$ is the mean annual frequency of exceeding a specific im value (known also as the hazard curve). $P(DS|im)$ is the fragility function for the asset which is located at the point where risk is being calculated and $f(RC|DS)$ is the consequence function for the asset in question.

In the case of human exposure, we can calculate the mean annual rate $\lambda(L > l)$ of exceeding loss value l as following:

$$\lambda(L > l) = \sum_{n=1}^2 \int_0^{\infty} \mathbf{1}(L > l|X_F) P(F|im) |d\lambda(IM > im)| \quad (17)$$

Where $L = X_F N$ is number of casualties equal to the product of the Bernoulli variable X_F and the total number of people N .

For physical exposure, it is customary to collapse the damage state to derive the *vulnerability curve*. The vulnerability curve is defined as:

$$P(L > l|im = x) = G_{L|IM}(l|im) = \sum_{n=1}^{N_{DS}} P(L > l|DS) P(DS|im) \quad (18)$$

Vulnerability is defined as the CCDF (complementary cumulative distribution function, see [Lecture Note 2](#) and [Lecture Note 3](#)). Note that we are using the fact that for one asset, the total direct economic loss L is equal to the repair cost RC . You can note that vulnerability is obtained through convolution of the consequence $P(RC > l|DS)$ and the fragility $P(DS|im)$. Therefore, we can re-write the risk integral in Equation 16 above as a convolution of hazard and vulnerability:

$$\lambda(L > l) = \int_0^{\infty} P(L > l|im) |d\lambda(IM > im)| \quad (19)$$

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In the case of human exposure, we can calculate the *fatality rate* as the mean annual rate $\lambda(F)$:

$$\lambda(F) = \int_0^\infty P(F|im) |d\lambda(IM > im)| \quad (20)$$

Where $P(F|im)$ is the fatality function. Note that the expressions in Equations 19 and 20 in general do not have a closed form solution and can be resolved using numerical integration.

Considering epistemic uncertainties in the hazard curve

If the hazard curve is furnished with epistemic uncertainty estimation (which is usually the case), we can propagate this uncertainty all the way to the risk. More specifically, imagine that the epistemic uncertainty is expressed with Lognormal epistemic uncertainty ε_{UH} with unit median and logarithmic standard deviation equal to $\sigma_{lnUH} = \beta_{UH}$ (see [Lecture Note 7](#) for Lognormal Distribution and its properties). We can calculate the loss curve $\lambda(L > l|\varepsilon_{UH})$ for a given random deviation ε_{UH} from the median hazard curve as following:

$$\lambda(L > l|\varepsilon_{UH}) = \int_0^\infty P(L > l|im) |d(\hat{\lambda}(IM > im) \cdot \varepsilon_{UH})| \quad (21)$$

Where $\hat{\lambda}(IM > im)$ is the median hazard curve and $\hat{\lambda}(IM > im) \cdot \varepsilon_{UH}$ is a plausible realisation of the hazard curve. This means, we can calculate a loss curve corresponding to a specific deviation from the hazard curve. Repeating this procedure for a representative set of ε_{UH} values, we can obtain the mean and the percentiles for the loss curve due to epistemic uncertainty in the hazard curve. The procedure is the same for human exposure, so we do not repeat it here. Note that we can also consider epistemic uncertainties in the fragility, consequence and fatality models and propagate them in the same way to the level of loss curve.

A closed form and analytic solution for the loss curve

We can obtain closed-form analytic solutions for the loss curve, based on some simplifying assumptions. It has been shown (e.g., Cornell et al. 2002, Jalayer et al. 2020), that a closed and analytic solution for the risk integral can be obtained if the hazard curve is approximated with a power-law expression and the fragility, vulnerability, and fatality functions are represented with Lognormal distributions (explained in detail below, you can find a related proof in [Lecture Note 7](#)). Let us assume that the hazard curve can be represented as $\lambda(IM > im) = k_o im^{-k}$ where k_o and k are intercept and slope parameters, respectively.

The case of human exposure: Let us assume that the Fatality function is fully described as the Lognormal cumulative distribution function for the tsunami intensity value (e.g. flow depth) that is life-threatening denoted as IM_F . This Lognormal distribution will be fully defined knowing the median η_{IM_F} and logarithmic standard deviation $\sigma_{lnIM_F} = \beta_{IM_F}$. Note that the Fatality function $P(F|im) = P(IM_F \leq im) = F_{IM_F}(im)$; in other words, the probability of loss of life for a given level of intensity im can also be interpreted as the probability that life-threatening intensity IM_F is less than im . This happens to be also the definition of the cumulative distribution function (CDF) for random variable IM_F denoted as $F_{IM_F}(im)$. Therefore, the risk integral can be written as follows:

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$$\begin{aligned}\lambda(F) &= \int_0^\infty F_{IM_F}(im) |d\lambda(IM > im)| = \int_0^\infty \lambda(IM > im) dF_{IM_F}(im) \\ &= \int_0^\infty \lambda(IM > im) f_{IM_F}(im) dim = \int_0^\infty k_o im^{-k} f_{IM_F}(im) dim\end{aligned}\quad (22)$$

We have used the integration by parts to re-arrange the above expression. Moreover, we have used the relation between PDF and CDF (PDF is the derivative of CDF, see [Lecture Note 3](#)). The risk integral in Equation 22 reduces to the expected value of a lognormal variable raised to a power. As it is shown in [Lecture Note 7](#), the expected value of a Lognormal random variable raised to a power, has an analytical solution:

$$\lambda(F) = \int_0^\infty k_o im^{-k} f_{IM_F}(im) dim = E(k_o im^{-k}) = k_o \eta_{IM_F}^{-k} e^{\frac{1}{2} \beta_{IM_F}^2} = \lambda_H(IM_F) e^{\frac{1}{2} \beta_{IM_F}^2} \quad (23)$$

This means that the fatality rate defined as the mean annual rate of loss of life, will be equal to hazard value at the median life-threatening intensity IM_F (as a first order approximation) times exponential of the square of the logarithmic standard deviation of IM_F divided by two. It should be noted that in the log-log scale, the tsunami hazard curve usually has a significant curvature and is not a line. However, it has been shown that if we approximate the hazard curve with the tangent line at the median life-threatening intensity IM_F , we will obtain very reasonable estimates of risk (see Jalayer et al 2020).

The case of direct economic losses: In the case of economic losses, we assume that the vulnerability CCDF is described by a Lognormal distribution fully defined by knowing the median η_{IM_L} and logarithmic standard deviation $\sigma_{ln IM_L} = \beta_{IM_L}$, where IM_L is the value of intensity corresponding to 50% probability that loss L exceeds specific value l . The risk integral in Equation 18 can be re-arranged as follows using integration by parts:

$$\begin{aligned}\lambda(L > l) &= \int_0^\infty P(L > l|im) |d\lambda(IM > im)| = \int_0^\infty G_{L|IM}(l|im) |d\lambda(IM > im)| \\ &= \int_0^\infty \lambda(IM > im) |dG_{L|IM}(l|im)| = \int_0^\infty k_o im^{-k} |dG_{L|IM}(l|im)| \\ &= \int_0^\infty k_o im^{-k} f_{L|IM}(l|im) dim\end{aligned}\quad (24)$$

We have used the relation between PDF and CCDF (PDF is the minus derivative of the CCDF, see [Lecture Note 3](#)). The risk integral in Equation 24 reduces to the expected value of a lognormal variable raised to a power. As it is shown in [Lecture Note 7](#), the expected value of a Lognormal random variable raised to a power, has an analytical solution:

$$\begin{aligned}\lambda(L > l) &= \int_0^\infty k_o im^{-k} f_{L|IM}(l|im) dim = E(k_o im^{-k}) = k_o \eta_{IM_L}^{-k} e^{\frac{1}{2} \beta_{IM_L}^2} \\ &= \lambda_H(IM_L) e^{\frac{1}{2} \beta_{IM_L}^2}\end{aligned}\quad (25)$$

This mean that the loss curve, defined as the mean annual rate of exceeding a specific loss value, will be equal to hazard value at the intensity corresponding to 50% probability of exceeding economic loss l (as a first order approximation) times exponential of the square of

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the logarithmic standard deviation of IM_L divided by two. As mentioned above, it has been shown that if we approximate the hazard curve with the tangent line at η_{IM_L} , we will obtain very reasonable estimates of risk (see Jalayer et al 2020). To calculate the loss curve, the closed-form solution in Eq. 24 will need to be calculated for all the loss levels l . This is not very convenient, since the best place to fit the tangent line is going to depend on value l .

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